

Banking Technology Lab

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Banking technology Lab



Definition

As part of the Banking Technology lab, students will develop multiple small applications related to the banking domain. Integrating these applications, which will be built and deployed in the Cloud using Azure App Service.

The outcome of this lab would be a fully functional banking application that is hosted on the cloud.

The lab would comprise the following Modules:

- Product and Pricing
- Fixed Deposit Calculator
- Fixed Deposit Module →
 - Account creation – Fixed Deposit (Liability Class)
 - Interest calculation, capitalization, and interest payouts
 - Maturity Calculations
 - Pre-Maturity Withdrawal
- Reports – Customer communication, Operational reports, Compliance reports, etc.
- Customer → Account Portfolio in Customer module developed in Fintech. Integrate with the Fixed Deposit account module.

Technology Stack

- Development environment: Dev Containers,
- Docker and Kubernetes
- Java Technologies – Core Java, Java Script, Spring boot, Angular, HTML, CSS
- SQL on MYSQL
- Python

Learnings from Fintech lab to be put in use

- Apply the basic concepts learnt in Fintech Labs
- Apply the technology and architecture patterns learnt in Fintech Lab
- Apply the authorization pattern learnt in Fintech
- Use the Customer module and integrate with the Accounts module from Banking Technology Lab
- API's, DB, UI, Reports to be built as defined in Fintech Lab

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Learning from Banking Technology Lab

- How different modules can be developed independently and integrated
- Admin Function vs Bank User Functions
- FD Calculations and its variations(Simple and Compounding): building simulators
- Account Life cycles and processes: Fixed Deposit
 - Creation, Interest calculation, Payouts, Pre-processes, Maturity processing, Closure, etc.
- Time travel testing: Testing through different life cycles of an account
- Processes in Batch processing: Interest calculation, statement generation, Maturity processing, etc.
- Different transaction types, balance types for different product/account types
- Debit/Credit, Asset/ Liability Classification, and GL transactions and aggregation
- Customer communication, Bank operational communication, Compliance Communication, etc.

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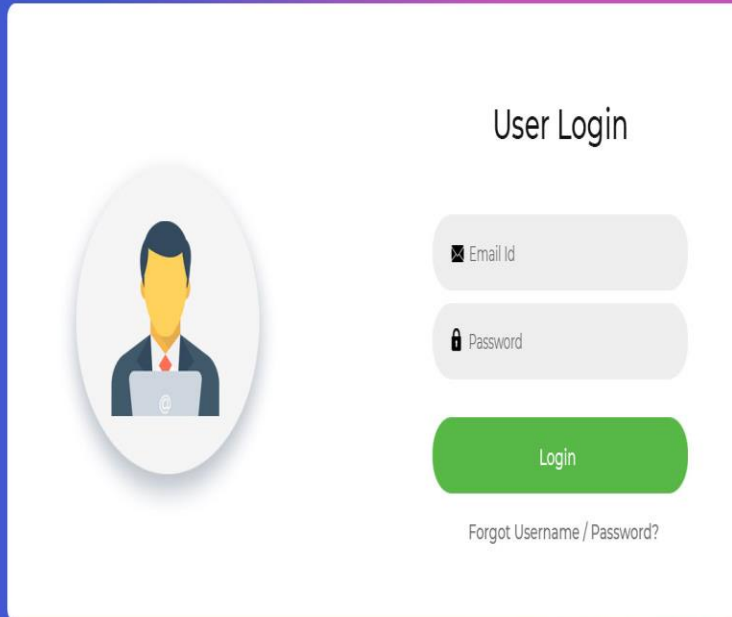


Learning from Banking Technology Lab

- Authorization and Authentication, JWT token, → Use open source like Bcrypt for authentication
- Double-byte character set, usage of Index, ISO standards, Standard formats like Amount format
- Caching, Hashing, Defining the rules and regulations
- API, Events, and File extract.
 - API Backward compatibility
 - Event publishing → Event – Driven API
 - File extract
- Batch Processing
- Logging and code Scanning
- Elasticity → Scaling up of Pods

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Modules – User Registration and Banking Application



The image shows a 'User Login' interface. On the left is a circular profile picture of a man in a suit. To the right are two input fields: 'Email Id' with an envelope icon and 'Password' with a lock icon. Below these is a green 'Login' button. At the bottom, there is a link that says 'Forgot Username / Password?'.

User Registration and Banking System → Admin function

- Application to register a user
 - Encryption and Decryption of the password using salt and hashing
 - Register the user in the Customer Module and use the customer ID as a User.
 - Multiple user ID options, e-mail, and phone number can be used instead of the customer ID.
 - Use of a strong password policy like minimum length of 8, with at least 1 number, 1 special character, and 1 Capital letter.
- Application to verify the user and allow access to the system
- Both the User-ID and password must be entered to log in successfully.
- Integrated system consisting of Customer, Product and Pricing, Fixed Deposit Calculator, Fixed Deposit, Report Modules
- SSO for all applications, each application should be able to log in and work independently
- Set a limit to auto logoff if the page is idle for a predefined time. E.g. 5 minutes

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Modules – Product and Pricing



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Product and Pricing → Admin function

Define products offered by banks like CASA, TD, Unsecured Loan, etc.

Defining business rules, guiding behaviors, roles, relationships, etc for the products

A generic way of designing an application to accommodate a varied set of values.

Functionality with a few of the business rules - minimum age, Term for Deposits, Deposit and Withdrawal Amount limits, etc.

Different Life cycle statuses for different product types

Different interest types(Debit for Loans/Credit for Deposits) for different product types.

Different rates based on customer attributes like Senior citizen, Employee, etc.

Different types of Fees – Periodic, one-time, event-based fee(e.g. Withdrawal from ATM), etc.

Different balance types, transaction types for different products.

Understand how business rules are defined and built.

Defaulting of product definition on accounts, resetting of values on the account, etc.E.g. Term, amount, etc., on an FD account.

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Modules – Fixed Deposit Calculator



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Fixed Deposit Interest Rate Excel Calculator

(Change only the Red field)

Capital Invested	Interest Rate	Type of Interest	Compounding Frequency	Years	Compound Interest Amount	Total Maturity Amount with Interest
100000	10	Simple Interest Rate	NA	5	50000	150000
100000	10	Compound Interest Rate	Yearly	5	61051	161051
100000	10	Compound Interest Rate	Half Yearly	5	62889	162889
100000	10	Compound Interest Rate	Quarterly	5	63862	163862
100000	10	Compound Interest Rate	Monthly	5	64531	164531

FD Calculator → Bank user function

Define a module that takes Amount, tenure, interest rate, start date and calculates the Maturity values of amount and date for a Fixed Deposit.

Interest calculation when

- Compounding at different frequencies
- Paid out on a regular basis, like Monthly, quarterly, etc.

Calculate APY

Integrate with the product module to fetch the required details from the product when the product code is provided.

Cache the rate information and retain it till logged out.

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Modules – Fixed deposit Account Module



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Fixed Deposit Account Module → Bank_User function

- This module needs to be integrated with the following:
 - Product to get the product details
 - FD calculator Module
 - Customer Module
- Store the default value from the product and create the account → . Calculate maturity values using the FD Calculator
- Change some values on account (from what is defined on the product) store the details; calculate maturity values using the FD Calculator
- Based on customer type, application of different rates → Students, Senior citizens, Employees of the bank etc. have different rates
- Transaction on the account → Initial deposit – Credit transaction, Early withdrawal leading to reduced interest, Withdrawal on Maturity, interest payouts.
- Batch for interest Calculation, Statements, Capitalization and payout of Interest, Maturity processing
- Customer communication through (Email, Alert) FDR during account creation, transaction alerts, interest processing, Maturity Notice, etc.
- Operational reports → Total Liability, Monthly growth, etc.
- Compliance reports

We can demo the below to the students:

- Product and pricing design, product and pricing defined for TD
- Time travel of the account → Creation, Interest Processing, Pre-closure, Maturity, renewal, etc.
- Interest processing on the account
- Maturity processing on the account
- Tax implications and compliance, etc.
- Internal accounts – double bookkeeping
- GL Transactions
- Business Event-based design
- Event publishing and usage of events
- Reports and Extracts